Relation as Category

An Epistemological Investigation

Harald Høffding

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Despite the great expansion of the horizon that Copernicanism gave impetus to, natural science had held fast to the concepts of absolute place and absolute motion, which had their justification only within the old worldview with its bounded and fixed frame. Galileo's and Newton's absolute space replaced the space bounded by the eighth sphere. But there came a time when the concept of relation here too made its consequences felt, and not only with respect to the relations among objects, but also with respect to the relation between objects and the knowing subject.

It was the first proposition of kinematics, and thereby of natural science, that was first examined from the standpoint of the concept of relation. Neumann (*Ueber die Prinzipien der Galilei-Newton'schen Theorie*, 1870) drew attention to the fact that the law of inertia, according to which the velocity and direction of a body change only through external influence, necessarily presupposes in each individual case the specification of the particular place in relation to which velocity and direction are preserved when no external influence is exerted. Neumann could find no place that could lay claim to being taken as the starting point for an absolute coordinate system, but held that one must hypothetically assume a center in order for the concept of absolute motion to retain its meaning. Mach, who at roughly the same time had arrived by his own path at the insight that the law of inertia rested on the presupposition of a determinate coordinate system, did not find Neumann's solution satisfying. He demanded that the concept of relation be carried through. The dogmatic conception of the law of inertia, in his view, only falls away when one becomes clear that in itself any body in the world can be used as a starting point, and that in practice in each individual case we use one body or another as the presupposed fixed starting point for the coordinates in relation to which the velocity and direction of a motion are determined (*Die Erhaltung der Arbeit*, 1872). This is the path that has in fact always been taken, only within the old worldview, which rested precisely on the presupposition of an absolute center and an absolutely fixed boundary-sphere of the world, no choice was required. Mach lays great weight throughout on the historical presuppositions on which the research of each period rests, but then shows how these presuppositions vary from age to age, and that none of them can lay claim to absolute necessity and intelligibility. What matters here always comes down to what French physicists call *points de repère*.[1]

An even more radical application of the concept of relation appeared at the beginning of the twentieth century. What is of interest in this connection is that it has proved necessary to take special account of the apprehending subject and its standpoint. Copernicus had already pressed this point as regards the apprehension of the motions of celestial bodies, and Neumann and Mach as regards the velocity

[1] Cf. e.g. Painlevé (*Mécanique*, p. 399) [in the collective work *De la Méthode dans les Sciences*, 1909]: "… les mouvements absolus, c'est-à-dire au fond les mouvements convenablement repérés." [i.e., absolute motions are, at bottom, motions suitably referred to reference points.]

and direction of motion. Einstein[2] has now sought to show that what holds for space and motion holds also for time. That this was not discovered earlier is, according to Einstein, to be explained by the fact that attention was paid only to velocities that were very small in relation to the velocity of light. The final temporal measure becomes, for modern physics, the velocity of light, whereas for Aristotle, and even for the Copernicans, it was drawn from the motions of celestial bodies. (Bruno's doctrine of the relativity of the concept of time rests on this measure. See above.) On the basis of the electromagnetic theory of physical phenomena, for which the mechanical conception of nature stands as only a special case, Einstein seeks to show that there can be demonstrated circumstances under which two observers, of whom one is attached to fixed and the other to moving axes, but both at rest relative to their own axes, will not arrive at the same apprehension of time: what for one observer stands as a relation of simultaneity between two events, will for the other stand as a relation of succession. A temporal specification acquires meaning only when it is stated in relation to which coordinate system it holds, and in particular whether that system is itself relatively at rest or in motion.

If one were to hold that an illusion must be present in one or the other observer, the answer from the Einsteinian standpoint is that each observer has his own system of measurement, which is just as legitimate as another's, though it is of course perfectly possible for one observer to attempt to place himself at another's standpoint and reason from it.[3]

The facts and inferences on which the Einsteinian theory rests are still a matter of dispute, and views stand sharply opposed to one another. In such a dispute the philosopher as such cannot intervene, and philosophy has time to wait. Whether physics can really demonstrate situations in which what stands as simultaneity for one observer stands for another observer as successive, physicists must determine among themselves. If this must be answered with yes, we stand before the significant result that natural science has once again, as with the Copernican hypothesis, by its own path come to see the necessity of not merely holding to the content of observations, but of taking account of the observer's presuppositions. In an interesting discussion

[2] *Über die spezielle und die allgemeine Relativitätstheorie* (1916).

[3] Langevin, in *Bulletin de la Société française de Philosophie*, 11 October 1911. — It must presumably be assumed that the observer who can place himself at another observer's standpoint knows sufficient physics.

in the *Aristotelian Society* (1919), Whitehead impressed upon us the significance of what he aptly called "the percipient event,"[4] and Nicholson declared that among the data of ultimate character there must also be counted "the essential framework of our modes of perception."

Against the Einsteinian hypothesis, as against earlier demonstrations of the epistemological significance of the concept of relation,[5] it has been urged that there can only be one truth in each individual case, and that it would contradict the principle of non-contradiction if two observers, of whom one declares simultaneous what the other declares successive, were both to be right.[6] But there is no reason to fear for the unity of truth, even if Einstein should be right. The real truth lies in this case in the necessity, physically demonstrated, that different observers under certain determinate conditions arrive at different apprehensions of time. Each individual observer apprehends time as he must under his conditions. But if an observer is also a physicist, he will (assuming the correctness of Einstein's theory) be able to explain to us why another observer arrives at a different apprehension than he himself does. (Cf. Newton's and Leibniz's emphasis on the lawfulness of motion as the expression of its objectivity. Above, p. 66.) Naturally, the physical explanation of the different observers' different apprehensions of time itself rests on certain presuppositions, presupposes a coordinate system from which the coordinate systems of the various observers are determined, and here there is the possibility of a new discussion. When this discussion comes to an end, it is because no standpoints can be demonstrated that are still more fundamental. Kant long since maintained that all necessity is hypothetical. But it is necessity nonetheless — or rather: precisely because of this it is necessity. A necessity without presuppositions we do not know.

If the new theory of relativity wins continued confirmation, what comes to hand will not merely be a result of physical research that is interesting in and of itself. This theory also contains, as Einstein himself has declared, a spur to continued inquiry. If it is the case that there is the possibility of different systems of relations in the

[4]See also his *Principles of Natural Knowledge* (1919), p. 68. — Whitehead's conception differs from Einstein's, however, in that "the percipient event" for him is only one factor alongside many others, and not, as in Einstein, of essential significance for all measurement of time and space. Cf. on this point Haldane: *The Reign of Relativity* (1921), pp. 69 f.; 107 ff.

[5]Cf. *Den menneskelige Tanke* [The Human Mind], p. 304.

[6]Kroman, in *Tidsskrift for Fysik*, XIV, p. 15.

apprehension of an object, this constitutes precisely an invitation to discover all the particular relations that lie at the basis in each individual case, and inquiry is thus led ever further rather than closing off in dogmatic fashion, as if the presuppositions one had hitherto worked with were the only possible ones. There are, from an epistemological point of view, two interests that attach to the concept of relation. One is that light is cast on the nature and conditions of our knowledge; the other is that new tasks are indicated, a *plus ultra* appears, in that every thought points beyond itself. —

Campanella said in his apology for Galileo that if Galileo were right, one would have to philosophize in a new way. But if Einstein is right, one need not therefore philosophize in a new way.[7] Even from a purely philosophical point of view, as the foregoing presentation has shown, attention was long since drawn to the significance of the concept of relation for knowledge and worldview. But if Einstein is right, there now lies before us a new, unexpected, but significant experience in that respect. —

Through the necessity of taking account of the knowing subject, and through the possibility of a conflict between the each-for-itself necessary apprehensions of different knowing subjects, the newer natural science — in Einstein as in Copernicus — offers an analogy to what asserts itself powerfully in the domain of the human sciences, as I shall now attempt to show.

[7] Einstein himself agrees with this. In his preface to Lucien Fabre's book *Les théories d'Einstein* (1921) he says: "The theory of relativity neither can nor will provide a world-system, but only indicate a limitation to which the laws of nature are subject." That it is not merely a matter of a limit, we have seen above.

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